

Please hold on for a moment while we enter your participant ID.

ID:

1. I feel that the predictions provided by the satellite image classifier are reliable.

☐ Strongly Disagree ☐ Disagree ☐ Neither Agree nor Disagree ☐ Agree ☐ Strongly Agree

2. I would trust the retrained satellite image classifier more if it were updated based on the test set I created.

☐ Strongly Disagree ☐ Disagree ☐ Neither Agree nor Disagree ☐ Agree ☐ Strongly Agree

3. I would trust the system if it was deployed in a real-world application.

☐ Strongly Disagree ☐ Disagree ☐ Neither Agree nor Disagree ☐ Agree ☐ Strongly Agree

4. Viewing the disagreements between my expected labels and the predicted labels of the classifier made me understand more the weaknesses of the system.

☐ Strongly Disagree ☐ Disagree ☐ Neither Agree nor Disagree ☐ Agree ☐ Strongly Agree

5. I was able to understand why the image classifier made mistakes.

☐ Strongly Disagree ☐ Disagree ☐ Neither Agree nor Disagree ☐ Agree ☐ Strongly Agree

6. The system reacts unpredictably.

☐

Strongly Disagree

☐

Disagree

☐

Neither Agree nor
Disagree

☐

Agree

☐

Strongly Agree

7. Overall, the image classifier is accurate.

☐

Strongly Disagree

☐

Disagree

☐

Neither Agree nor
Disagree

☐

Agree

☐

Strongly Agree

8. Can you select the 3 categories that were the most challenging for the satellite image classifier to recognize?

- ☐ Agricultural area
- ☐ Forest
- ☐ Industrial or commercial area
- ☐ Railway
- ☐ Residential area
- ☐ Road
- ☐ Public square or park
- ☐ Waterbody

9. Can you select the 3 categories that were the easiest for the satellite image classifier to recognize?

- ☐ Agricultural area
- ☐ Forest
- ☐ Industrial or commercial area
- ☐ Railway
- ☐ Residential area
- ☐ Road
- ☐ Public square or park
- ☐ Waterbody

10. Which of the following images is most likely to have the lowest confidence in the prediction?



Image 1:



Image 2:

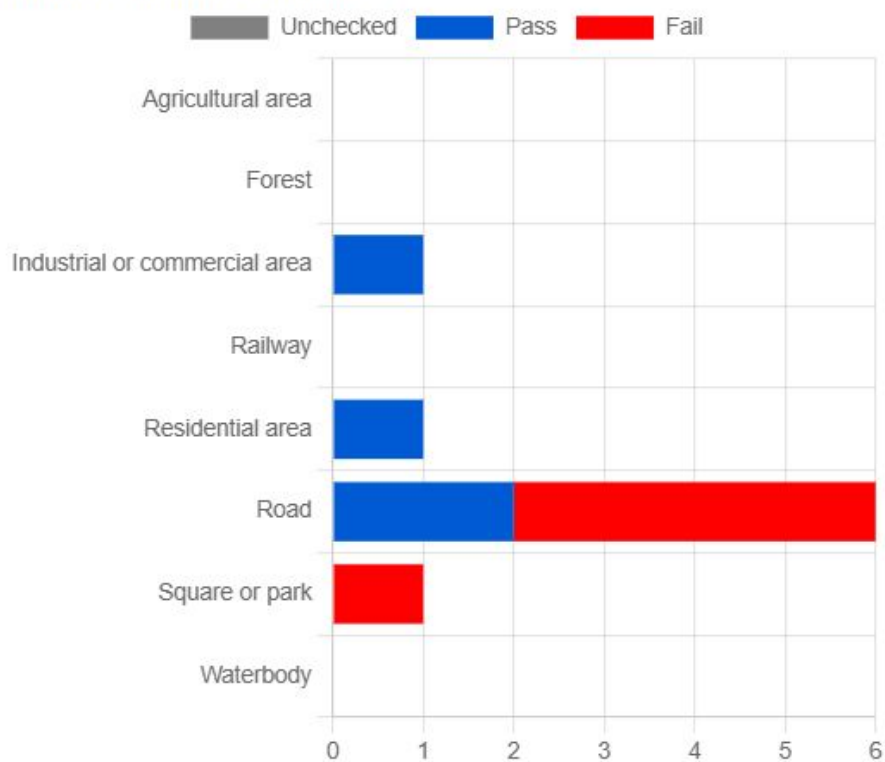


Image 3:

- ☐ Image 1
- ☐ Image 2
- ☐ Image 3
- ☐ I don't know

11. Consider the following bar plot to answer the question. How many instances of “Road” were correctly classified?

Total number of test cases: 9



- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ I don't know

12. After completing the experiment, I understand the reasoning behind the classifier's predictions.

☐

Strongly Disagree

☐

Disagree

☐

Neither Agree nor
Disagree

☐

Agree

☐

Strongly Agree

13. The predictions of the classifier align with my understanding of machine learning image classification.

☐

Strongly Disagree

☐

Disagree

☐

Neither Agree nor
Disagree

☐

Agree

☐

Strongly Agree

14. Which of the following best describes your strategy for creating your test set?

- ☐ I mainly selected challenging images for the classifier to uncover where the system fails.
- ☐ I mainly selected prototypical images for the classifier to assert where they system succeeds.
- ☐ I selected images that would produce a test set with equal number of correct and incorrect predictions.
- ☐ I selected images without a specific strategy.

15. Please briefly explain your strategy for creating your test set, if any:**16. Which of the following factors mostly influenced your choice of which classes to prioritize, as the testing progressed?**

- ☐ The bar chart
- ☐ The outcome (pass/fail) in the table
- ☐ The confidence value in the table
- ☐ The predicted label in the table
- ☐ My intuition or judgement
- ☐ Other (please specify):

17. Was there anything in the behavior of the classifier that you found surprising?

Yes (please explain what were your surprises):

☐☐

No

18. The bar chart showing the number of examples per class was useful to create my test set.

☐

Strongly Disagree

☐

Disagree

☐

Neither Agree nor
Disagree

☐

Agree

☐

Strongly Agree

19. The table information (pass/fail outcome, predicted label, confidence) was useful to create my test set.

☐

Strongly Disagree

☐

Disagree

☐

Neither Agree nor
Disagree

☐

Agree

☐

Strongly Agree

20. Is there any type of information that you feel would have been helpful in your testing, but was missing from the interface?

If yes, please explain.

If no, please leave the field blank.

21. How old are you?

22. Which gender do you identify with?

- ☐ Female
- ☐ Male
- ☐ Other
- ☐ Prefer not to say

23. How would you rate your expertise in machine learning?

- ☐ Neophyte: Never used it in practice.
- ☐ Beginner: Basic understanding, some theoretical knowledge.
- ☐ Intermediate: Applied it in practice occasionally, familiar with common algorithms.
- ☐ Advanced: Regularly use it in practice, strong understanding of various techniques and tools.
- ☐ Expert: Extensive experience, deep understanding, and possibly contribute to the field through research or teaching.

24. Have you taken any class related to machine learning in your studies?

- ☐ Yes
- ☐ No

25. Have you ever created a dataset to train a machine learning model?

- ☐ Yes
- ☐ No

26. Have you ever trained and tested a machine learning model on a dataset?

- ☐ Yes
☐ No

27. How familiar are you with the city of Munich and its surroundings?

- ☐ Not at all familiar ☐ Slightly familiar ☐ Familiar ☐ Very familiar ☐ Extremely familiar

Last Page

Thank you for completing this questionnaire!

We would like to thank you very much for helping us. Your contribution will be very valuable for our research.

Your answers were transmitted, you may close the browser window or tab now.